"A CYBERATTACK HAPPENS EVERY 39 SECONDS"

SECURITY AUTHENTICATION SYSTEM

INTRODUCTION

TEAM

CODE WARRIORS

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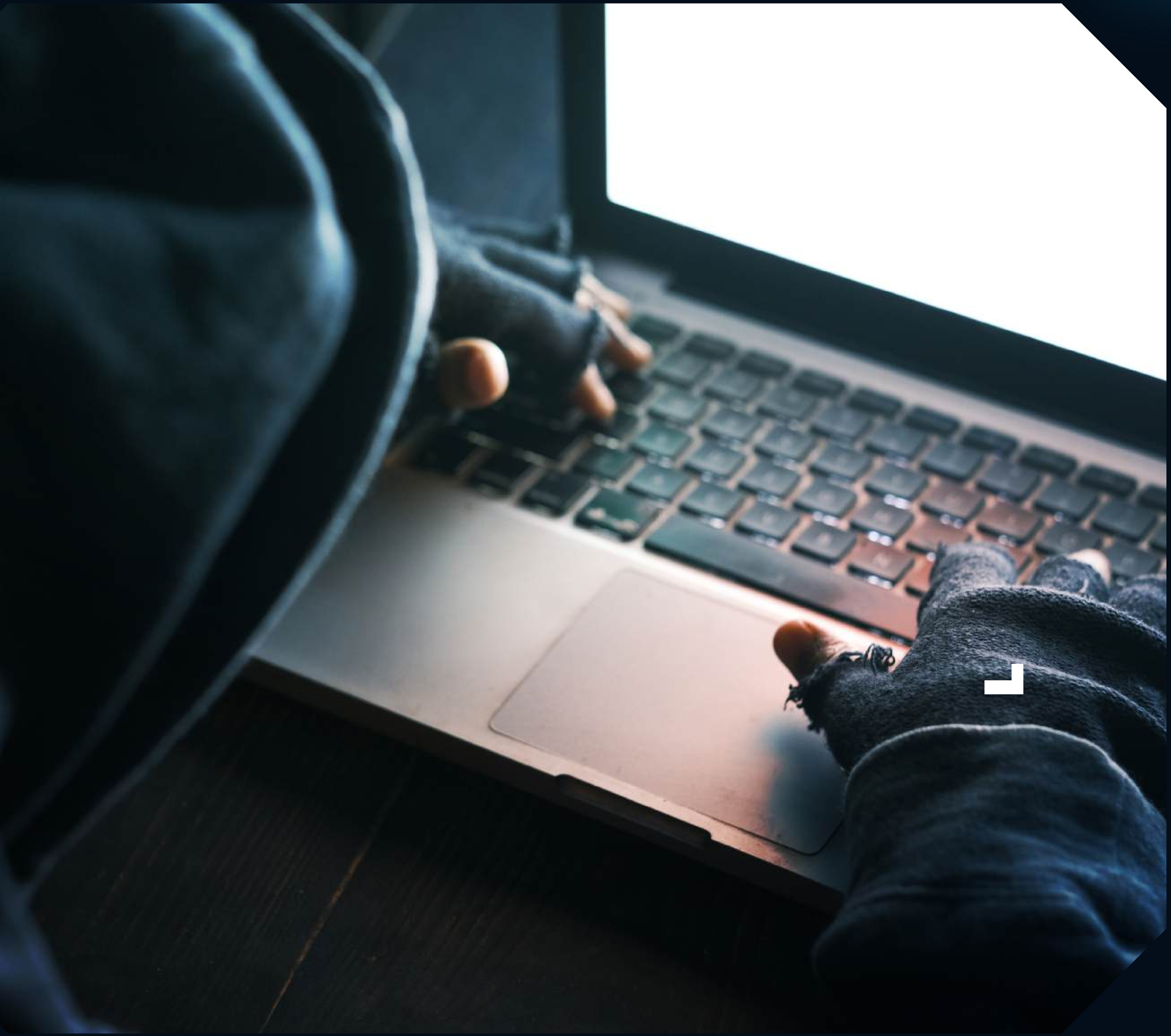
ECE 1ST YEAR

WHAT IS AUHTENTICATION?

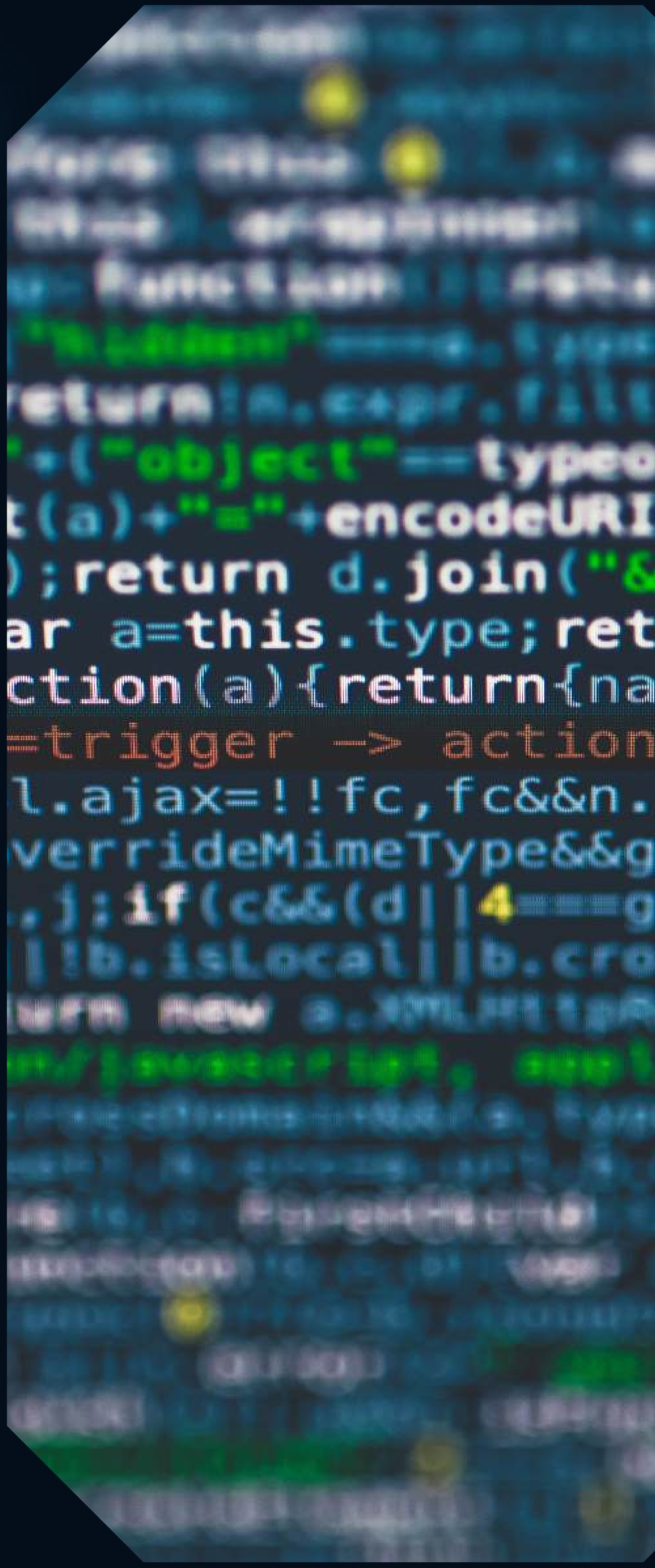


AUTHENTICATION IS THE PROCESS OF VERIFYING THE IDENTITY OF A USER, DEVICE, OR SYSTEM BEFORE GRANTING ACCESS TO RESOURCES, APPLICATIONS, OR SERVICES.

WHAT IS BIOMETRIC AUTHENTICATION?

BIOMETRIC AUTHENTICATION IS A SECURITY METHOD USED TO VERIFY A PERSON'S IDENTITY BASED ON THEIR UNIQUE BIOLOGICAL OR BEHAVIORAL CHARACTERISTICS. THESE CHARACTERISTICS ARE SPECIFIC TO EACH INDIVIDUAL AND CANNOT BE EASILY COPIED OR SHARED.

COMMON TYPES OF BIOMETRIC AUTHENTICATION INCLUDE FINGERPRINT RECOGNITION, FACIAL RECOGNITION, IRIS SCANNING, VOICE RECOGNITION, AND PALM PRINT RECOGNITION. DURING AUTHENTICATION, THE SYSTEM CAPTURES THE USER'S BIOMETRIC DATA AND COMPARES IT WITH THE STORED BIOMETRIC TEMPLATE TO CONFIRM THE USER'S IDENTITY.



WHY TO USE BIOMETRIC AUTHENTICATION?

BIOMETRIC AUTHENTICATION IS USED TO IMPROVE SECURITY AND PROVIDE A RELIABLE WAY TO VERIFY A PERSON'S IDENTITY. UNLIKE PASSWORDS OR PINS, BIOMETRIC TRAITS SUCH AS FINGERPRINTS, FACIAL FEATURES, OR IRIS PATTERNS ARE UNIQUE TO EACH INDIVIDUAL AND CANNOT BE EASILY FORGOTTEN, SHARED, OR STOLEN. USING BIOMETRIC AUTHENTICATION REDUCES THE RISK OF UNAUTHORIZED ACCESS AND IDENTITY FRAUD. IT ALSO MAKES THE AUTHENTICATION PROCESS FASTER AND MORE CONVENIENT FOR USERS BECAUSE THEY DO NOT NEED TO REMEMBER COMPLEX PASSWORDS.

USER AUTHENTICATION USING BEHAVIORAL BIOMETRICS:

- BEHAVIORAL BIOMETRICS IDENTIFIES USERS BASED ON THEIR BEHAVIOR PATTERNS WHILE USING A COMPUTER.

- TYPING PATTERN (KEYSTROKE DYNAMICS):
  - ANALYZES HOW A PERSON TYPES ON THE KEYBOARD.
  - MEASURES TYPING SPEED AND TIME BETWEEN KEY PRESSES.
  - EACH USER HAS A UNIQUE TYPING RHYTHM.

- MOUSE MOVEMENT ANALYSIS:
  - TRACKS HOW A USER MOVES THE MOUSE.
  - ANALYZES SPEED, DIRECTION, AND CLICKING BEHAVIOR.
  - EACH PERSON HAS A UNIQUE MOUSE USAGE PATTERN.

TECHNOLOGY STACK

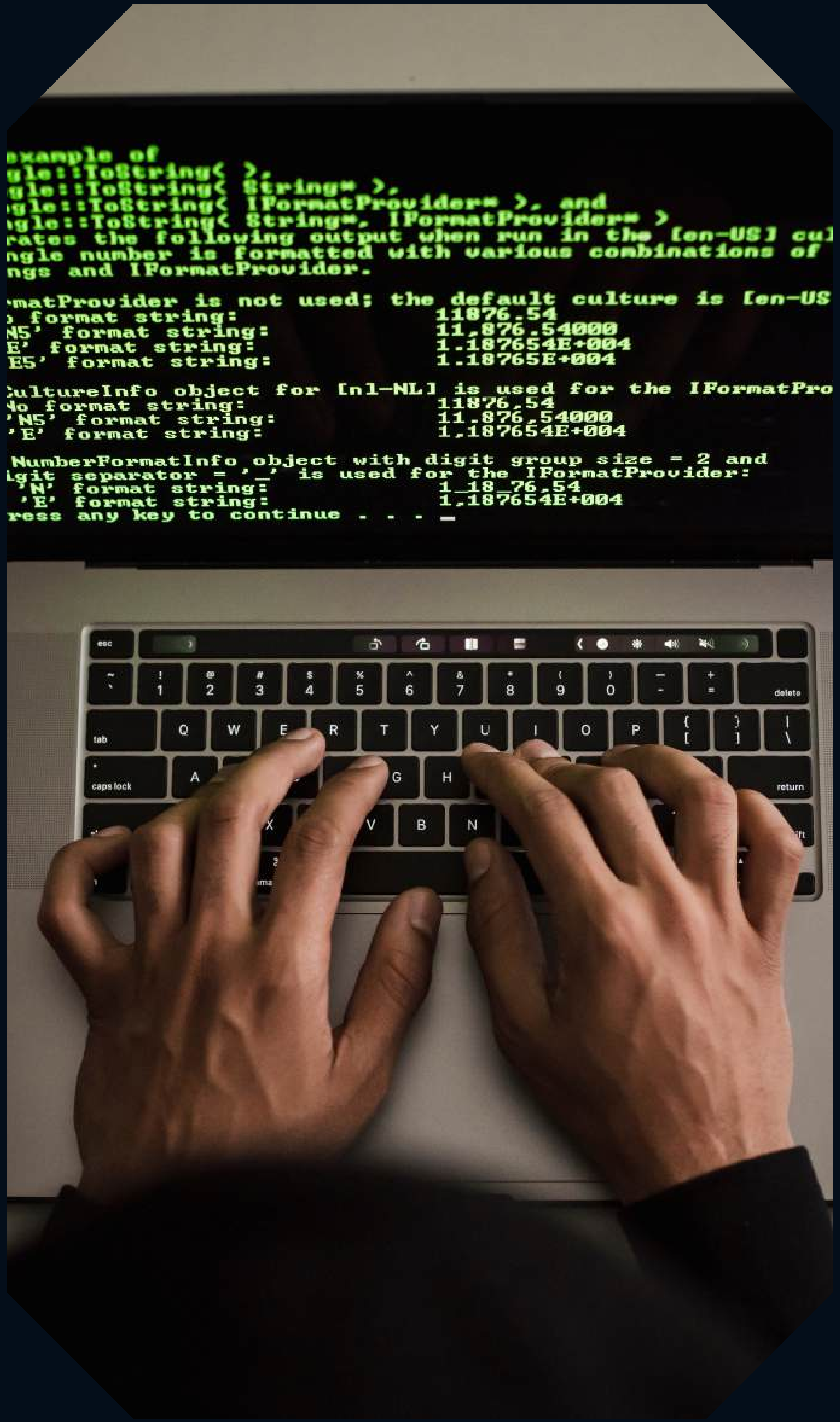
FRONTEND: HTML, CSS, JAVASCRIPT

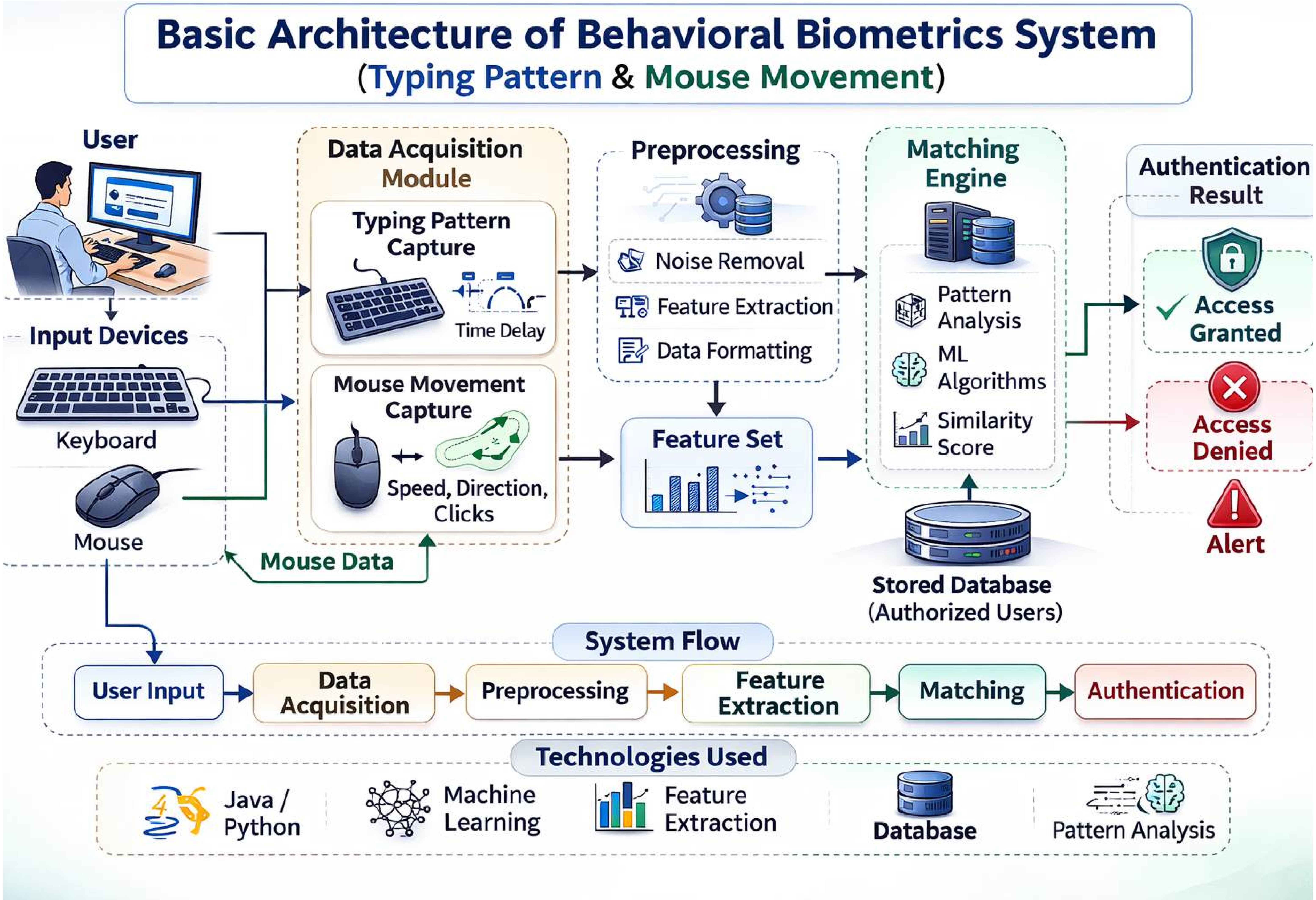
BACKEND: PYTHON

AI/ML: TENSORFLOW, OPENCV

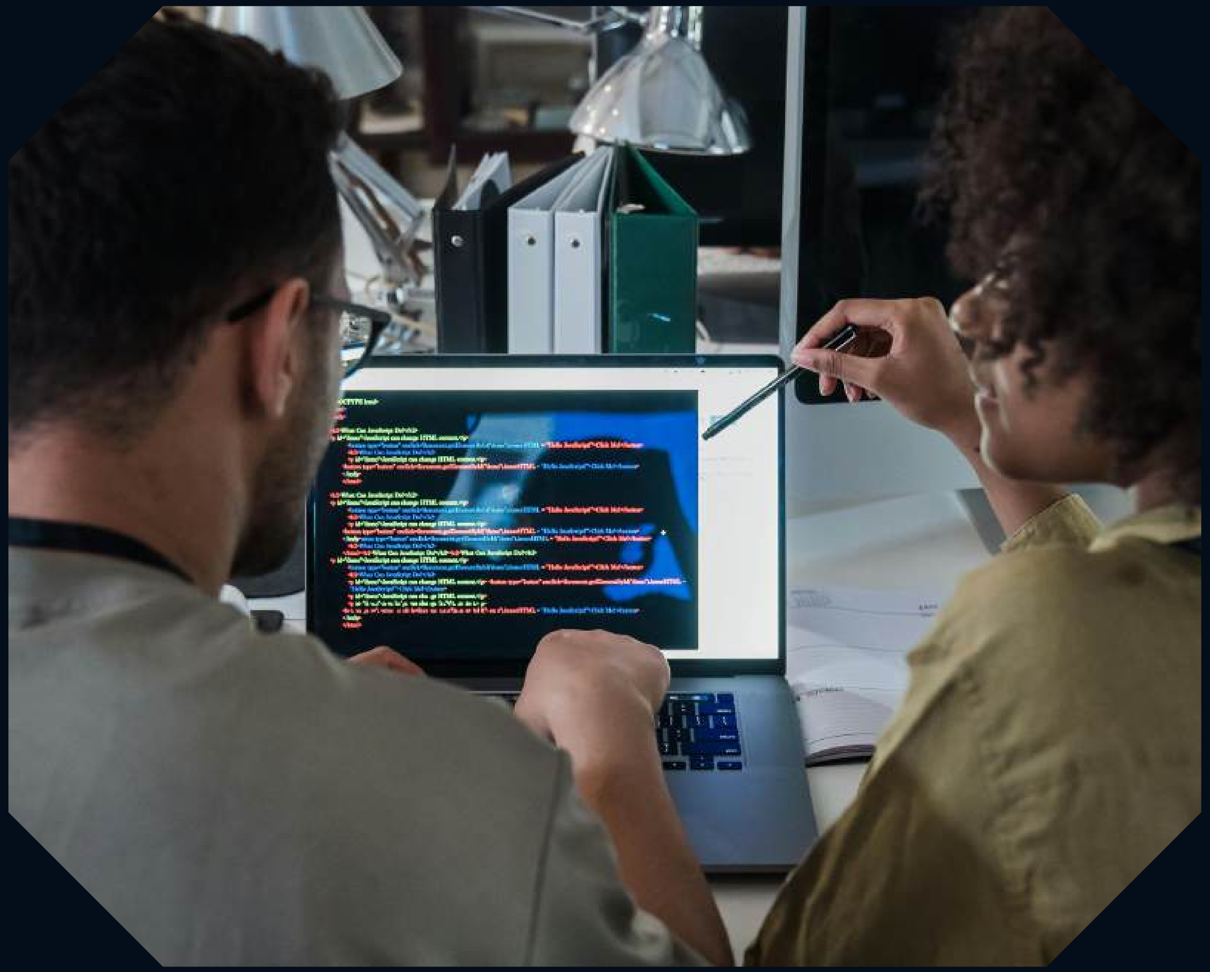
DATABASE: MYSQL / MONGODB

TOOLS: VS CODE, GIT




Basic Architecture of Behavioral Biometrics System
(Typing Pattern & Mouse Movement)
User
Input Devices
Keyboard
Mouse
Data Acquisition Module
Typing Pattern Capture
Time Delay
Mouse Movement Capture
Speed, Direction, Clicks
Mouse Data
Preprocessing
Noise Removal
Feature Extraction
Data Formatting
Feature Set
Matching Engine
Pattern Analysis
ML Algorithms
Similarity Score
Stored Database (Authorized Users)
Authentication Result
Access Granted
Access Denied
Alert
System Flow
User Input
Data Acquisition
Preprocessing
Feature Extraction
Matching
Authentication
Technologies Used
Java / Python
Machine Learning
Feature Extraction
Database
Pattern Analysis

EXPECTED OUTCOMES



- ACCURATE USER AUTHENTICATION BASED ON TYPING PATTERNS AND MOUSE MOVEMENT BEHAVIOR.
- IMPROVED SECURITY BY IDENTIFYING UNIQUE BEHAVIORAL PATTERNS OF EACH USER.
- ABILITY TO DETECT UNAUTHORIZED USERS AND SUSPICIOUS ACTIVITIES.
- CONTINUOUS MONITORING OF USER BEHAVIOR DURING SYSTEM USAGE.
- REDUCED RISK OF SPOOFING COMPARED TO TRADITIONAL AUTHENTICATION METHODS.